Class 5: Data Viz with ggplot

Barry (TSS: 911)

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Background

There are many *graphics systems* in R for making plots and figures. These include **"base R graphics"** like the `plot()` and add-on packages like **ggplot2**.

Let's compare how we make a simple graph with these two systems:

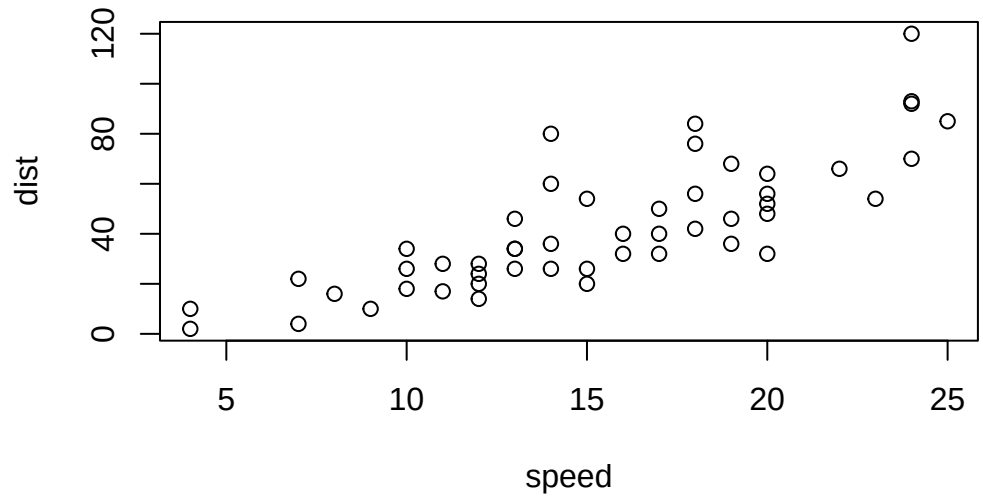
We can use the in-built `cars` dataset:

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

A base R plot:

```
plot(cars)
```



Now for **ggplot2**.

Before I can use ggplot2 I need to install it on my computer. To do this we can use the function `install.packages("ggplot2")`

> **N.B.** We never run an `install.packages()` in our quarto doc (we run it in our R console) as it would re-install the package every time I render my document.

Once installed we still need to load up the package into our R session. We do this with a `library()` call

```
library(ggplot2)
```

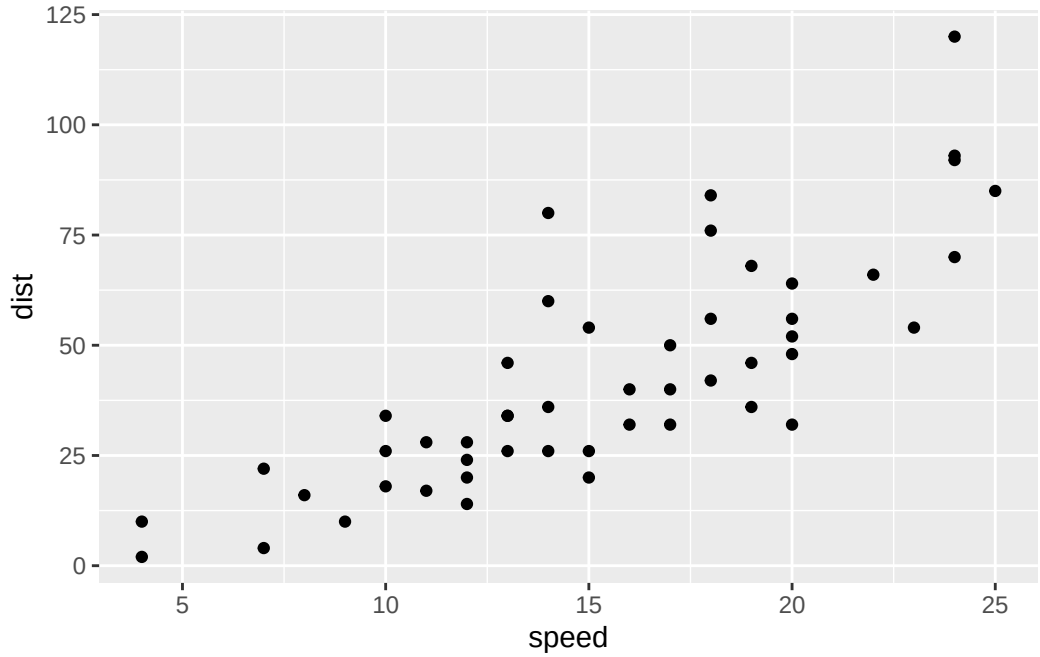
```
Warning: package 'ggplot2' was built under R version 4.4.3
```

```
ggplot(cars)
```

Every ggplot needs at least 3 inputs:

- the **data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot),
- the **geom** layer (how you want the plot drawn, points, lines etc.)

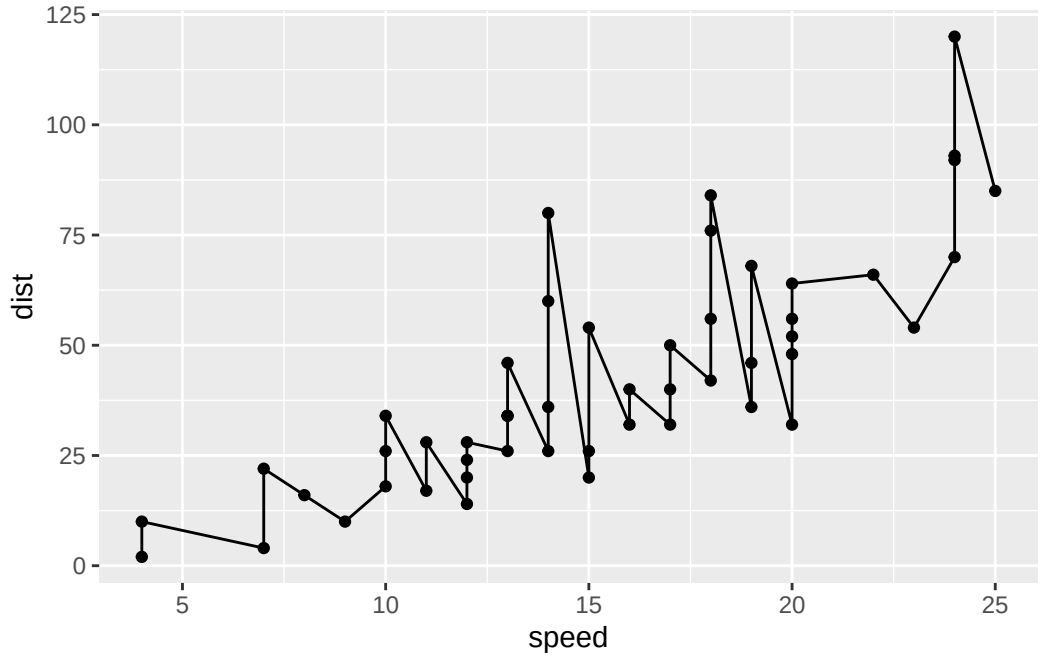
```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```



Lets add some features (more layers)

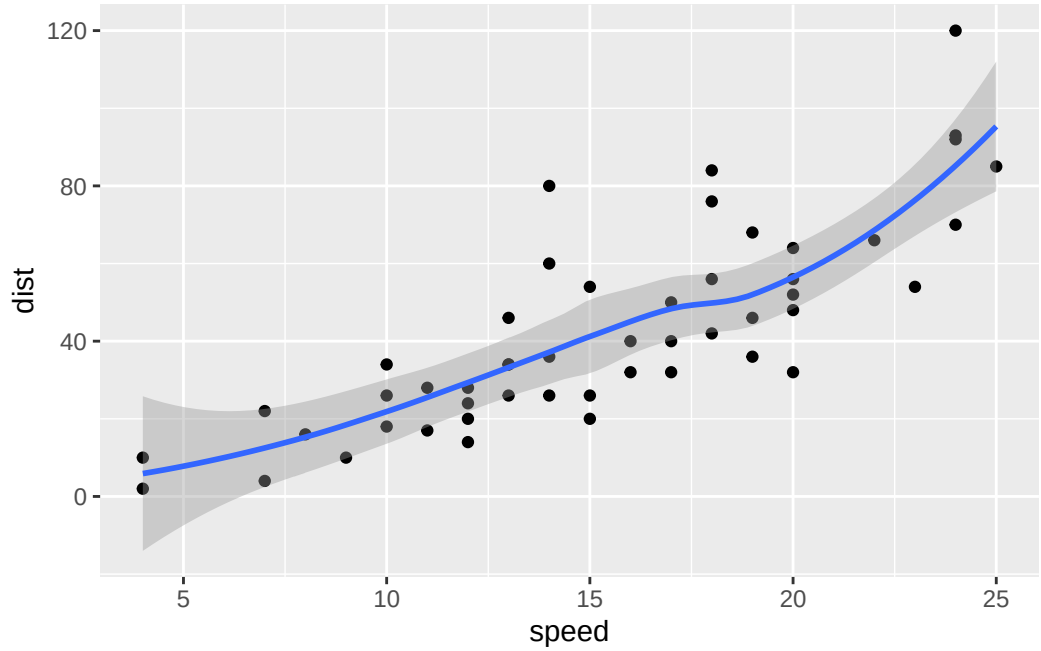
Let's add a trend line that shows the relationship between `speed` and `distance`:

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_line()
```




```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth()
```

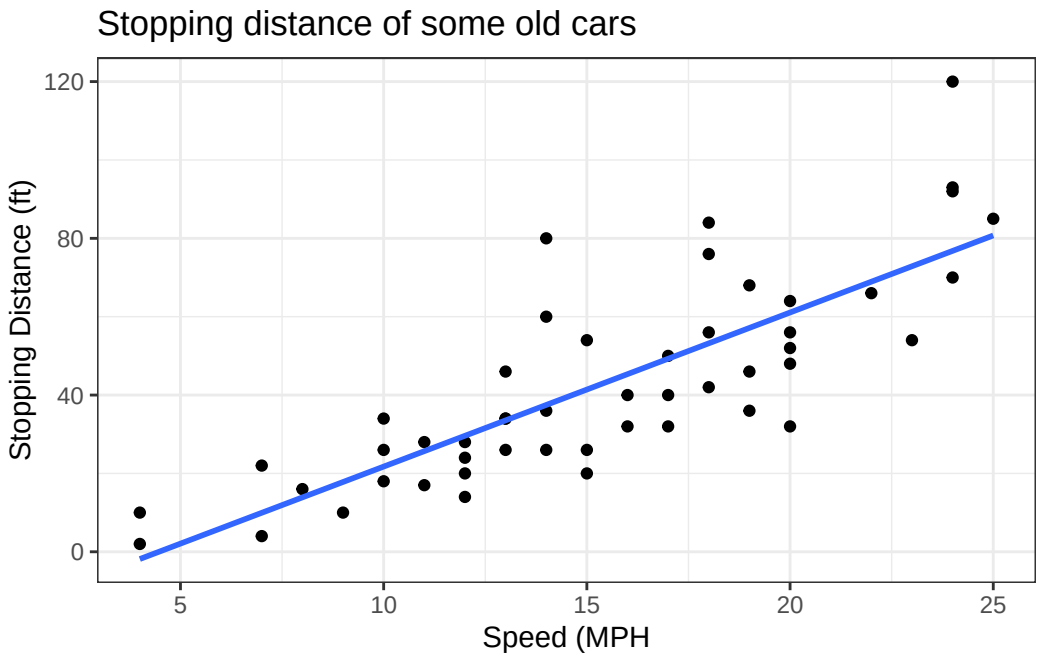
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



Add a theme, custom axis labels and plot title:

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw() +
  labs(x="Speed (MPH",
       y="Stopping Distance (ft)",
       title="Stopping distance of some old cars")
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Gene Expression Plot

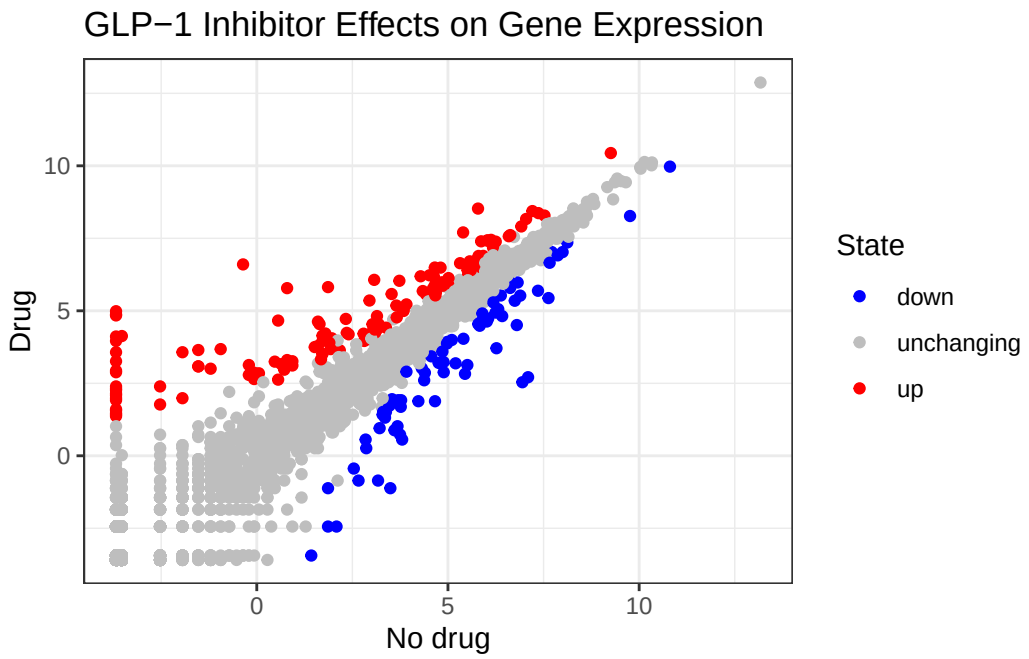
First we will read some expression data for the effects of a GLP-1 drug

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_colour_manual( values=c("blue","gray","red") ) +
  theme_bw() +
```

```
  labs(x="No drug",
       y="Drug",
       title="GLP-1 Inhibitor Effects on Gene Expression")
```



Q. Use the nrow() function to find out how many genes are in this dataset. What is your answer?

```
nrow(genes)
```

```
[1] 5196
```

Q. Use the table() function on the State column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

```
table(genes$State)
```

```
down unchanging         up
  72        4997        127
```

Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

```
n.total <- nrow(genes)
n.up    <- sum( genes$State == "up" )

round( (n.up/n.total*100), 2)
```

```
[1] 2.44
```

Going Further

Here we read the famous *gapmider* dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many entries are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

Q. How many different "country" values are in this dataset?

```
length( table(gapminder$country) )
```

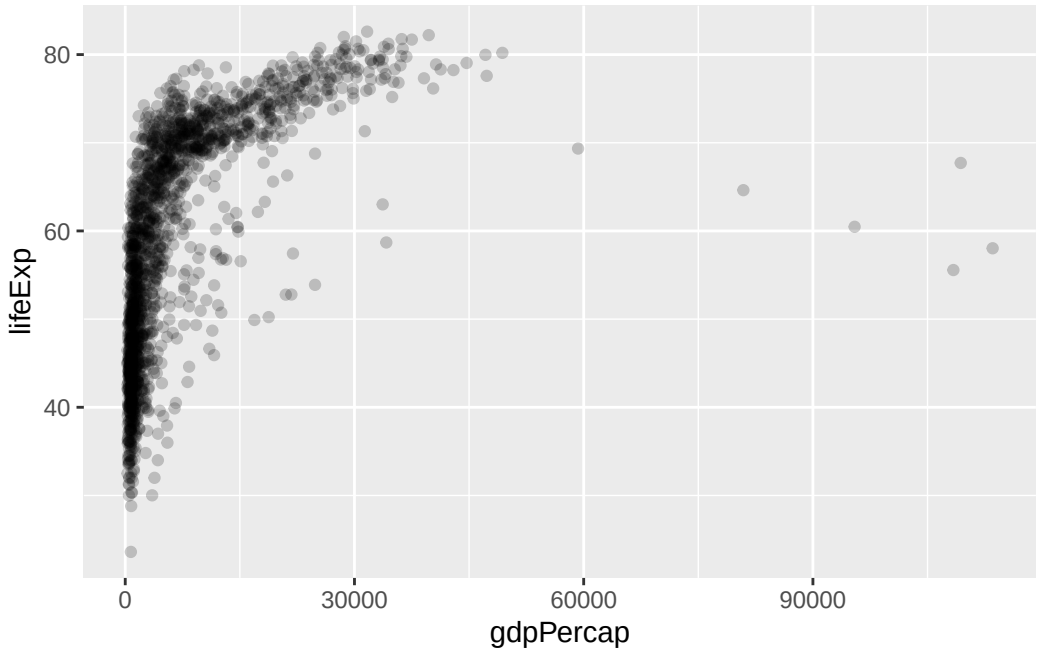
```
[1] 142
```

```
length( unique(gapminder$country) )
```

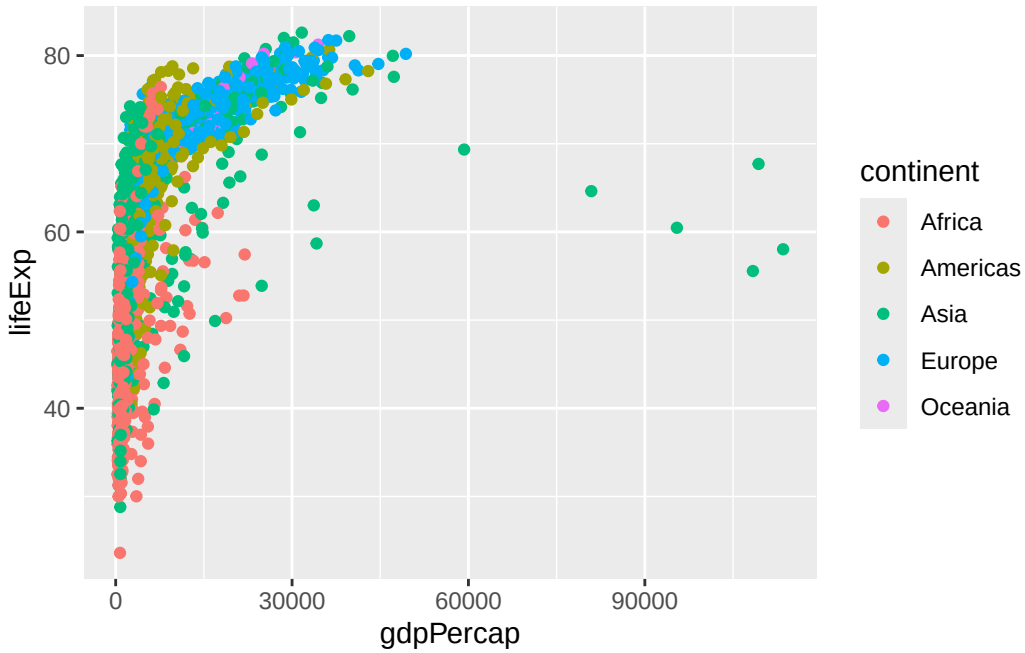
```
[1] 142
```

Let's make our first plot of `gdpPercap` vs `lifeExp`:

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp) +
  geom_point(alpha=0.2)
```

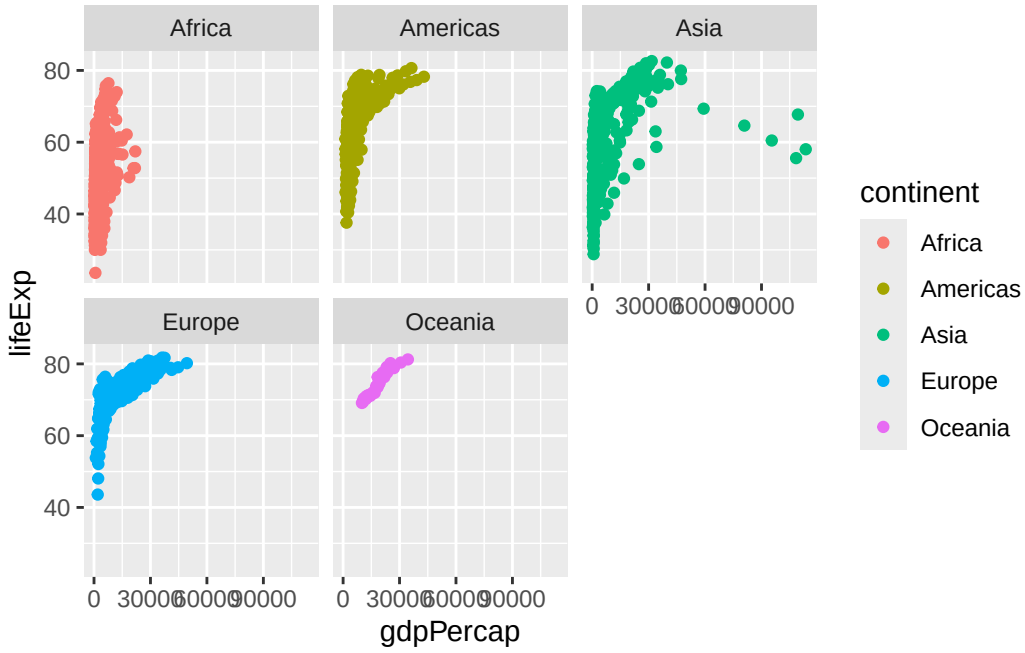



```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point()
```



One useful layer we can add to explore big datasets like this is called "facetting"

```
ggplot(gapminder) +
  aes(x=gdpPercap, y=lifeExp, col=continent) +
  geom_point() +
  facet_wrap(~continent)
```



Limit to 1977 and 2007

We can install and use the **dplyr** package to help us "filter" the full dataset down to a smaller subset that only includes the years we want.

First wee need to run `install.packages("dplyr")` and then load it with `library(dplyr)`.

```
library(dplyr)
```

```
Warning: package 'dplyr' was built under R version 4.4.3
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag
```

```
The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
filter(gapminder, year==2007, country=="Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 2007  78.885 4109086     40676
```

```r
filter(gapminder, year==1977, country=="Ireland")
```

```
  country continent year lifeExp     pop gdpPercap
1 Ireland    Europe 1977   72.03 3271900  11150.98
```

```r
#gapminder$year
```